Part Collection

This page records all the genetic parts and components used during our project.Based on our modular design for Small-Cell Lung Cancer (SCLC) therapy, our project utilizes a comprehensive library of genetic parts.

These are categorized into **Recycled Parts**—well-characterized components from the iGEM Registry optimized for our specific pulmonary environment—and **New Parts**—novel sequences derived from the latest scientific literature and patent data to provide specialized therapeutic functions.

Recycled parts

These parts were designed by adapting and combining existing iGEM parts that serve similar functions. Modifications were made using literature, to adjust them for our experimental setup.

Type	ID	Part name	Composite/Basic	Function and Description
coding	BBa_K2587001	$LuxI$	Basic	Together with the corresponding parts pLux and LuxR， it is able to induce expression of a gene of interest under the control of the pLux promoter.
coding	BBa_K2587002	$Plux$	Basic	Together with LuxI and LuxR , it is able to induce expression of a desired gene.
coding	BBa_K2587003	$luxR$	Basic	Together with the other two components (LuxI and Plux) it is able to induce expression of a downstream gene of interest. The amount of expression is based on cell population density.
coding	BBa_I0500	$ParaC$	Basic	The pBad/araC system is an arabinose-inducible, tightly regulated promoter in *E. coli* that drives gene expression upon L-arabinose addition
coding	BBa_K4156101	$PlldR$	Basic	The pLldR promoter is a lactate-inducible promoter that is repressed by the LldR protein in the absence of lactate but is activated by lactate to relieve repression and drive downstream gene expression

coding	BBa_R0040	$Ptet$	Basic	Promoter is constitutively ON and repressed by TetR. TetR repression is inhibited by the addition of tetracycline or its analog
coding	BBa_J23100	$PJ23100$	Basic	Ensures steady-state pre-armed expression of surface anchoring proteins and DLL3 nanobodies.
coding	BBa_C0040	$tetR$	Basic	Specifically binds to the *tet* operator within Ptet, enabling the negative feedback control shown in your circuit diagram.
coding	BBa_K554002	$HlyA$	Basic	The HlyA is a signal peptide found in the C-terminal signal sequence of alpha-hemolysin (HlyA). It is used to target proteins for secretion via the Type I secretion pathway of gram-negative bacteria.
coding	BBa_K2013021	$pelB$	Basic	The pelB signal peptide is used to direct the DLL3 nanobody to the periplasmic space or outer membrane surface, enabling active recognition of small cell lung cancer (SCLC).
coding	BBa_K1694010	$Lpp-OmpA$	Basic	Lpp-OmpA protein is the expression system of outer membrane protein, which consists of the 20 amino acids of signal sequence, the 9 N-terminal amino acids of the lipoprotein (Lpp) and the residual 46-159 amino acids of the OmpA.
coding	BBa_B0034	$StrongRBS$	Basic	High-efficiency ribosome binding site used for essential enzymes like $thyA$ or therapeutic payloads.
coding	BBa_B0030	$InternalRBS$	Basic	Positioned between $P30$ and VHH_{DLL3} to balance the expression ratio of anchoring and targeting proteins.
coding	BBa_B0015	Double Terminator	Basic	Ensures high termination efficiency and prevents transcriptional read-through between modules.

New parts

These parts were designed using sequences obtained from scientific literature rather than adapting existing iGEM parts. In some cases, existing iGEM parts feature similar constructs; when applicable, they are referenced on the Registry page.

Type	ID	Part name	Composite/ Basic	Sources	Function and Description
coding	To be assigned	$Lpp\text{-}OmpA\text{-}P30$	Basic	https://patents.google.com/patent/CN112979769B/zh	Fuses the P30 protein to Lpp-OmpA for mechanical anchoring of bacteria to SCLC cell surfaces.
coding	To be assigned	$pelB\text{-}VHH_{DLL3}$	Basic	https://www.guidetopharmacology.org/GRAC/LigandDisplayForward?tab=summary&ligandId=12969	DLL3-specific nanobody for active molecular recognition of small-cell lung cancer cells.
coding	To be assigned	$FuncPept$	Basic	Best Basic Part.pdf	This codon-optimized genetic sequence encodes an engineered peptide designed to induce the irreversible aggregation of the oncogenic transcription factor N-MYC within SCLC cells, thereby suppressing tumor progression.